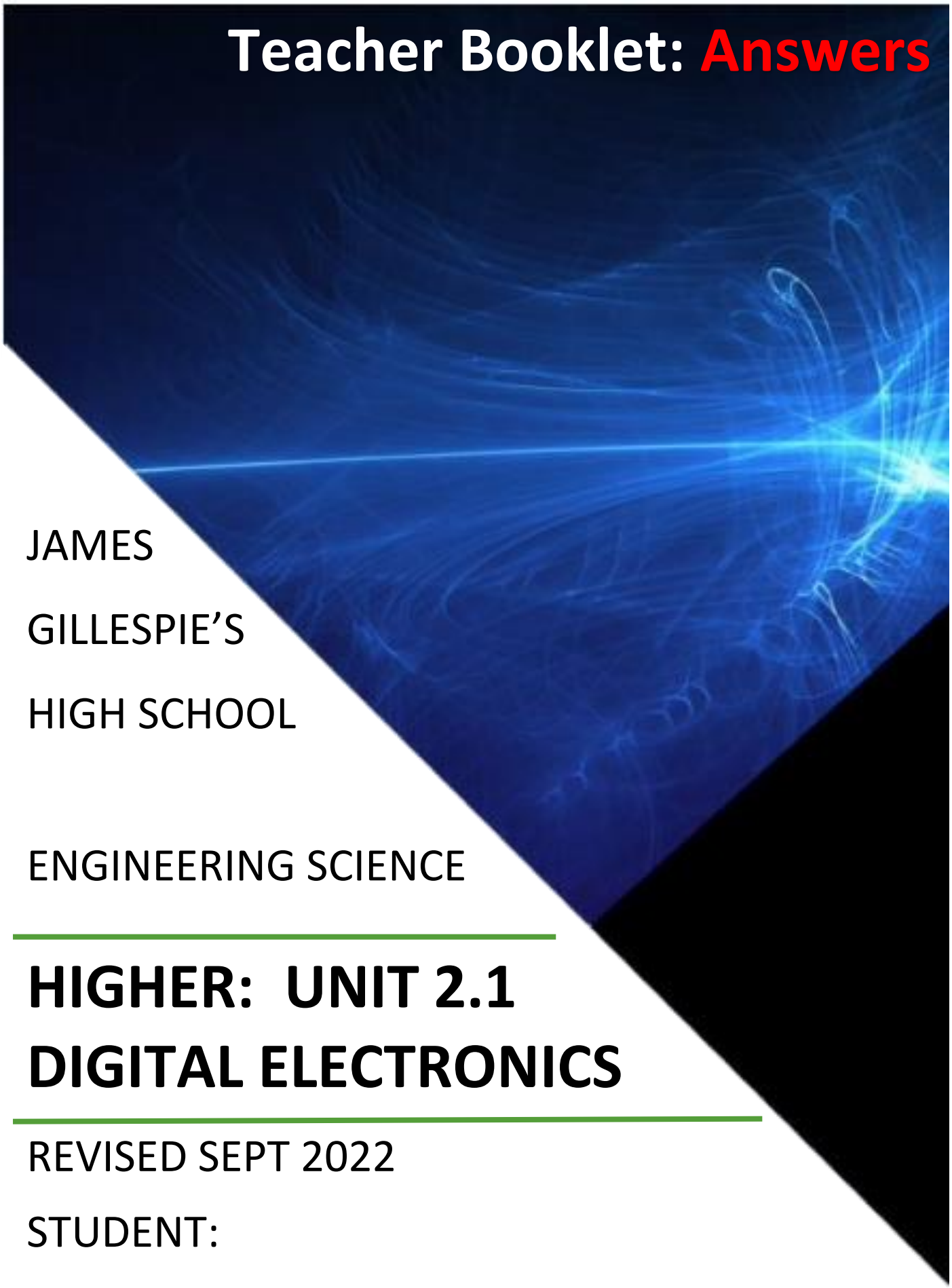
Teacher Booklet: Answers



JAMES

GILLESPIE'S

HIGH SCHOOL

ENGINEERING SCIENCE

HIGHER: UNIT 2.1 DIGITAL ELECTRONICS

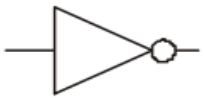
REVISED SEPT 2022

STUDENT:

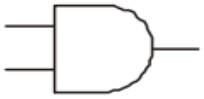
YEAR: TEACHER:

1

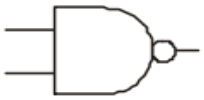
Students copy info below into jotters

NOT 

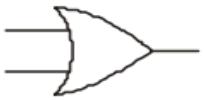
When a high signal inputs, a low signal outputs.

AND 

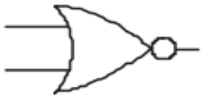
When a low signal inputs, a high signal outputs.

NAND 

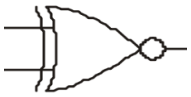
When both signals are high, output is high. Otherwise, output is low.

OR 

The opposite of an AND gate: When both signals are high, the output is low.

NOR 

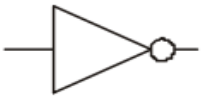
When one or both of the inputs is high, the output is high.

XOR 

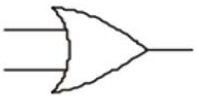
The opposite of an OR gate: When none of the inputs are high, the output is high.

2

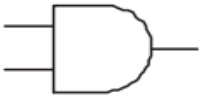
a) Draw a truth table for each of the logic gates shown on the previous page.

NOT 

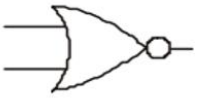
A	Z
0	1
1	0

OR 

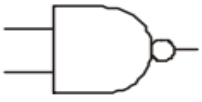
A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

AND 

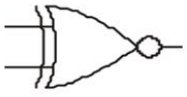
A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

NOR 

A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0

NAND 

A	B	Z
0	0	1
0	1	1
1	0	1
1	1	0

XOR 

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	0

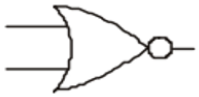
b) How many combinations of input states would there be for a 6 input system?

$2^6 = 64$ input states

c) Write down the 8 possible combinations of input states for a 3 input system.

A	B	C
0	0	0
0	1	0
0	0	1
0	1	1
1	0	0
1	1	0
1	0	1
1	1	1

d) To avoid accidents at times of poor visibility, a warning indicator in a car operates if the light level is too low (logic level 0) and the headlamps are switched off. Which logic gate would be needed for this operation?

NOR 

3

a) Why is the NOT gate sometimes called an 'INVERTER'

Because it inverts the signal it receives.

b) When is the output of an OR gate high?

When either input is on, or both.

c) When is the output of an AND gate high?

Only when both inputs are high

d) When is the output from a NAND gate high?

When either A or B is high, or both are low, but not when both are high (opposite of AND)

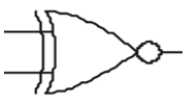
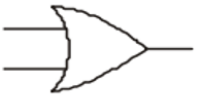
e) The truth table below shows the output conditions for the various combinations of input conditions for AND, NAND and OR gates. Complete the table and discuss how the output of the NAND gate compares with the output of the AND and OR gates.

Inputs	AND	NAND	OR
0 0	0	1	0
0 1	0	1	1
1 0	0	1	1
1 1	1	0	1

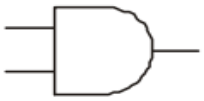
A NAND gate and an OR gate have the same number of high output states, but an OR gate is low when both inputs are low, and a NAND gate is low when both inputs are high. A NAND gate is the same as a combination of an AND and NOT gate, it inverts the AND outputs.

4

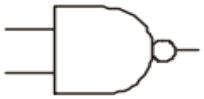
The paper can be fed through a computer printer either by pressing the button on the printer (line feed) or by sending a signal from the computer. State the logic gate which should be should be used for this operation

XOR  or OR 

5 The motor in a washing machine should not operate until a high signal is sent from the control program and the water level in the drum is high enough. State the logic gate which should be should be used for this operation

AND 

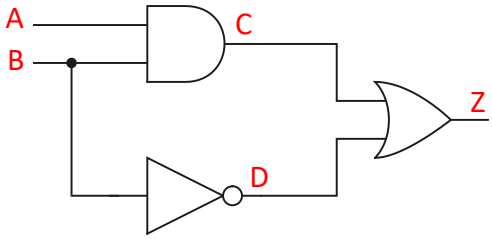
6 In the maternity unit of a hospital, the temperature and pulse rate of premature babies has to be continually monitored. A warning alarm should sound if either the temperature or the pulse rate of the baby falls too LOW.

NAND 

7

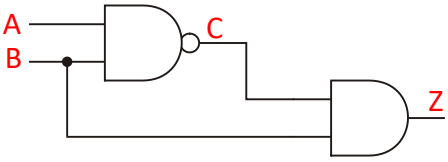
Complete a truth table for each of the combinations of gates shown below.

1.



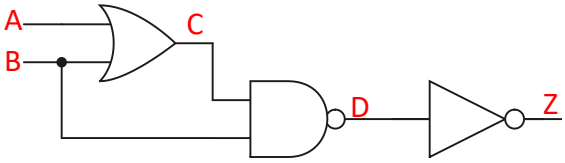
A	B	C	D	Z
0	0	0	1	1
0	1	0	0	0
1	0	0	1	1
1	1	1	0	1

2.



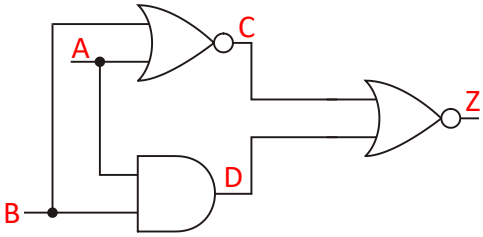
A	B	C	Z
0	0	1	0
0	1	1	1
1	0	1	0
1	1	0	0

3.



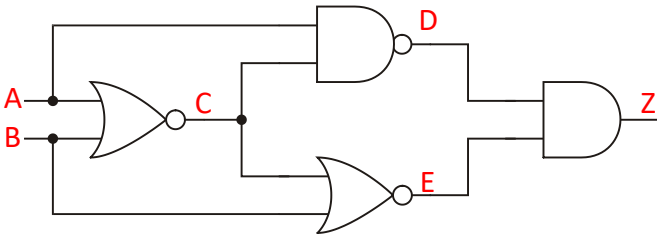
A	B	C	D	Z
0	0	0	1	0
0	1	1	0	1
1	0	1	1	0
1	1	1	0	1

4.



A	B	C	D	Z
0	0	1	0	0
0	1	0	0	1
1	0	0	0	1
1	1	0	1	0

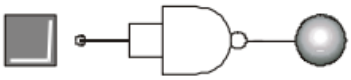
5.



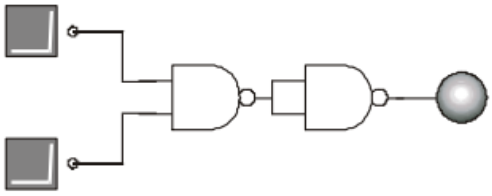
A	B	C	D	E	Z
0	0	1	1	0	0
0	1	0	1	0	0
1	0	0	1	1	1
1	1	0	1	0	0

NAND gates

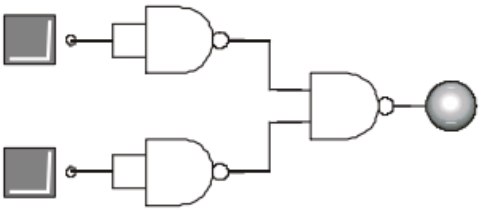
Students are to work out which NAND gate array represents which logic gate



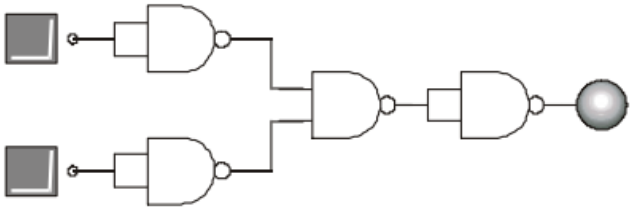
NOT gate



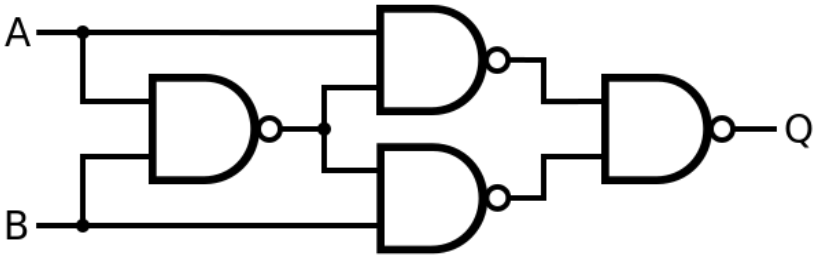
AND gate



OR gate



NOR gate

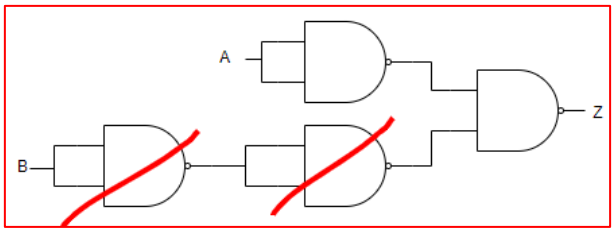


XOR gate

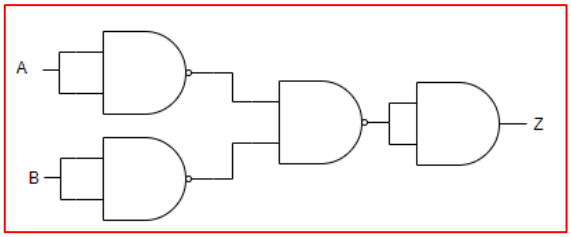
8

Students expected to give the answers below AND redraw the simplifaction.

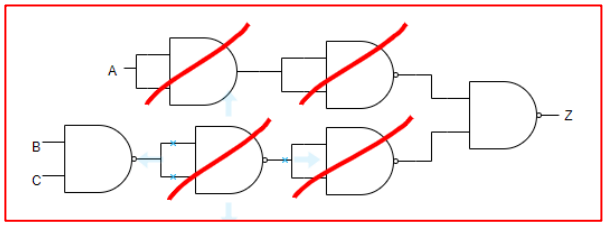
a)



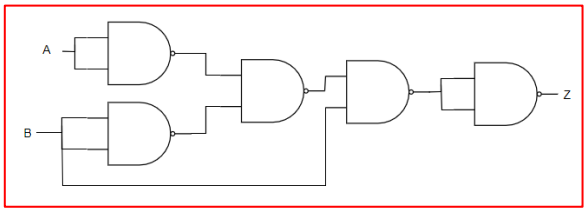
b)



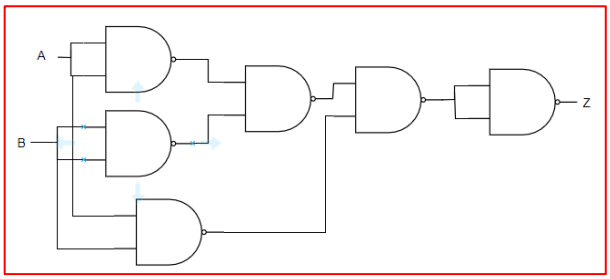
c)



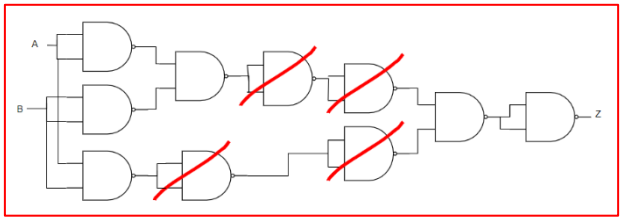
d)



e)



f)



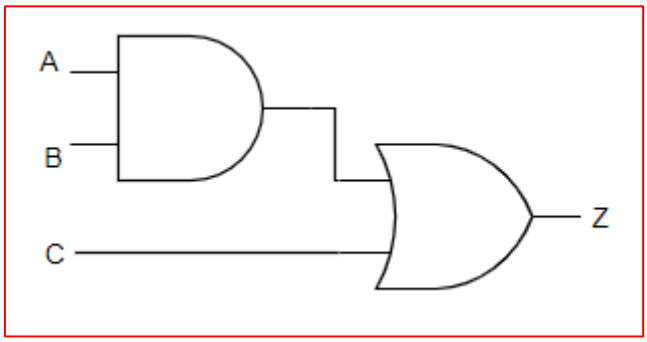
9

a) $A \cdot B = Z$
b) $\overline{A \cdot B} = Z$
c) $A + B = Z$
d) $\overline{A + B} = Z$
e) $\overline{A} = Z$
f) $A \cdot B \cdot C = Z$
g) $\overline{A \cdot B \cdot C} = Z$
h) $A + B + C + D = Z$
i) $\overline{A + B + C + D} = Z$

10

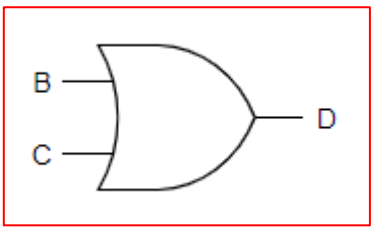
a) $(A + B) \cdot B = Z$
b) $(\overline{A \cdot B}) \cdot (A + B) = Z$
c) $(A + B) + (\overline{A \cdot B}) = Z$
d) $(A \cdot B) \cdot (C \cdot D) = Z$
e) $(\bar{A} \cdot B) + (A \cdot C) = Z$
f) $(\bar{A} \cdot B) + (A \cdot \bar{B}) = Z$

11

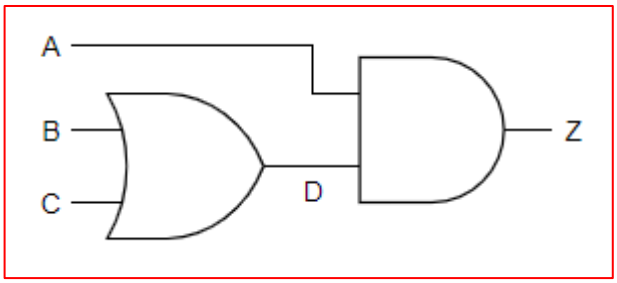


12

a) Deal with inputs B & C first then introduce input A for solution (b)



b)



c) $A \cdot (B + C) = Z$

13

a) $\overline{(A + B) \cdot (\overline{B + C})} = Z$ OR $(\overline{A + B}) \cdot (B + C) = Z$

b) $(A \cdot B) + C = Z$

A	B	C	Z
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Truth Table (b)

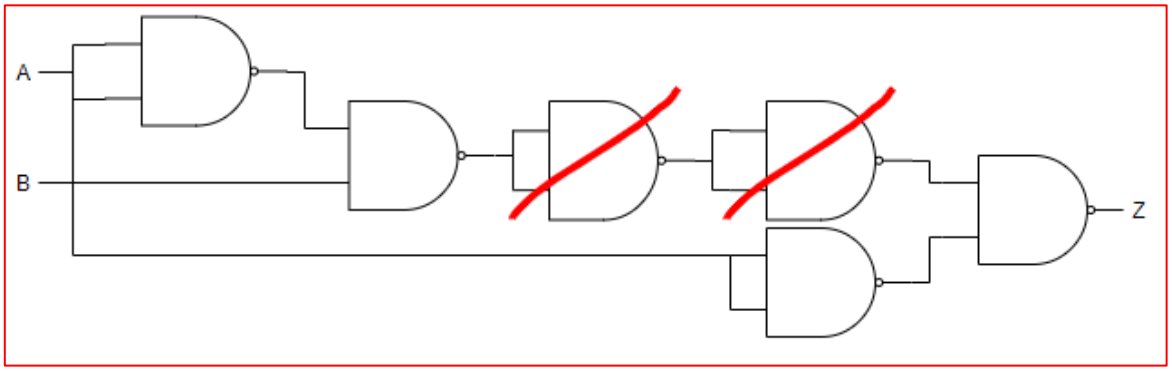
A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

14

a) $(\bar{A} \,.\, B) + A = Z$ AND $(A + B) = Z$

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

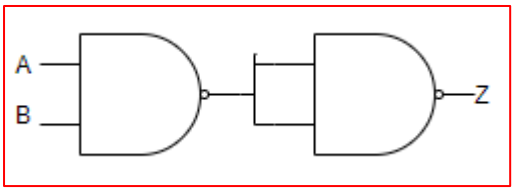
A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1



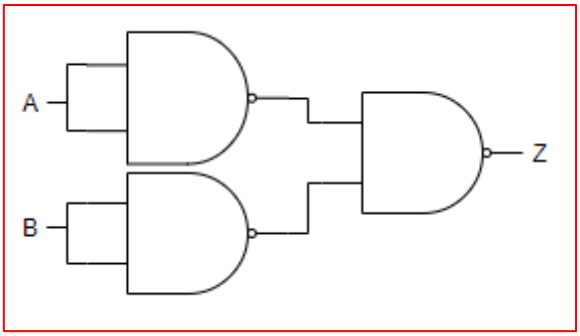
b) $A \cdot B = Z$ AND $A \cdot B = Z$

A	B	M	Z
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

A	B	M	N	Z
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1



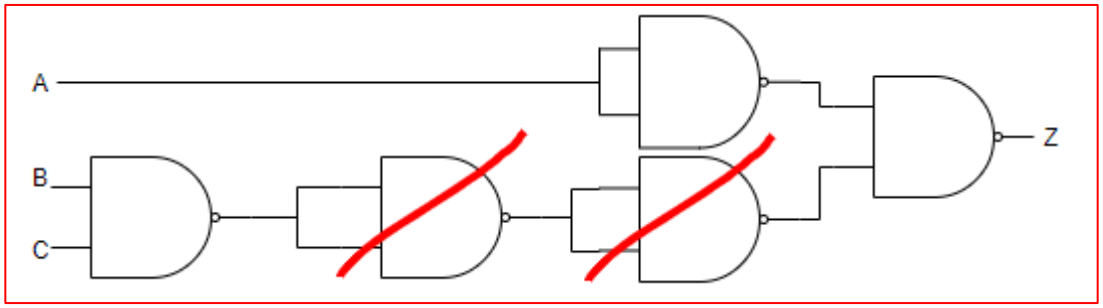
OR



c) $A + (B \cdot C) = Z$ AND $(A + B) \cdot (A + C) = Z$ or $A + (B \cdot C) = Z$

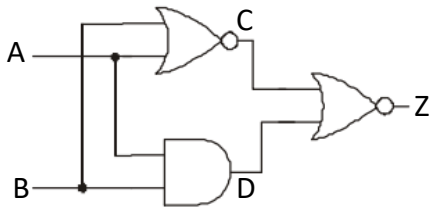
A	B	C	Z
0	0	0	0
0	1	0	0
0	0	1	0
0	1	1	1
1	0	0	1
1	1	0	1
1	0	1	1
1	1	1	1

A	B	C	Z
0	0	0	0
0	1	0	0
0	0	1	0
0	1	1	1
1	0	0	1
1	1	0	1
1	0	1	1
1	1	1	1



15

a)



$C = \overline{(A + B)}$

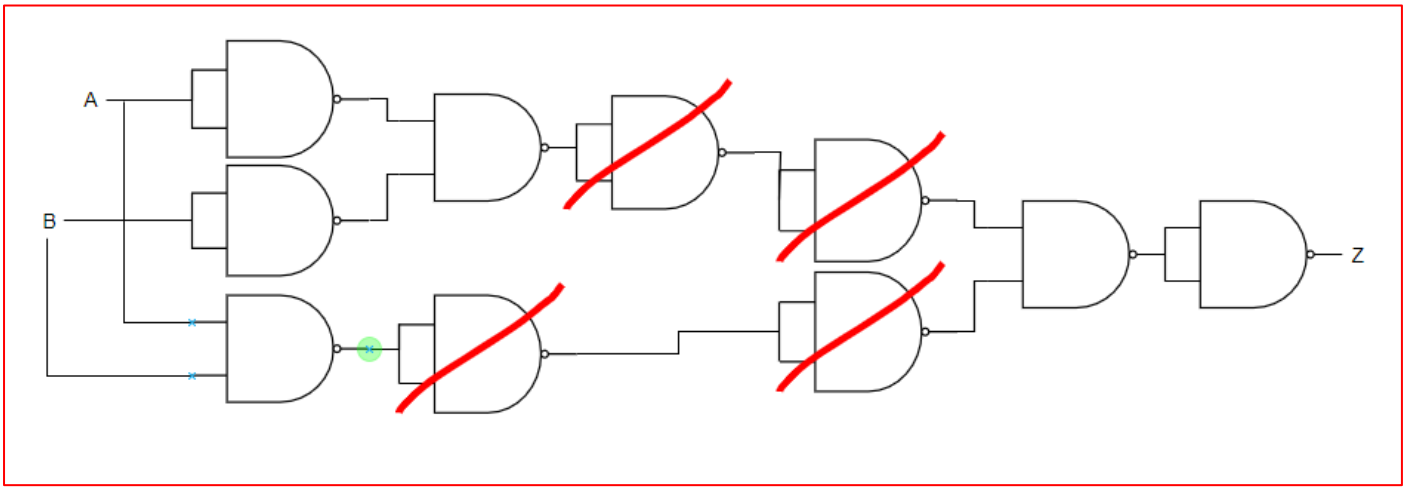
$D = A \cdot B$

$Z = \overline{(C + D)}$

$Z = \overline{\overline{(A + B)} + (A \cdot B)}$

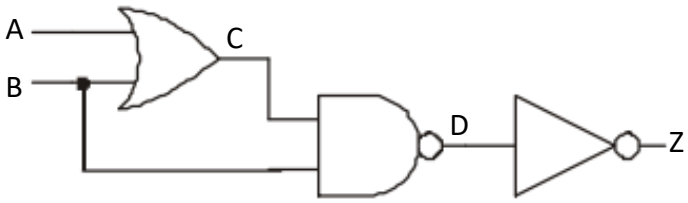
$Z = (A + B) + \overline{(A \cdot B)}$

A	B	C	D	Z
0	0	1	0	0
0	1	0	1	1
1	0	0	1	1
1	1	0	1	0



Students are expected to redraw the simplification of this logic diagram.

b)



$C = (A + B)$

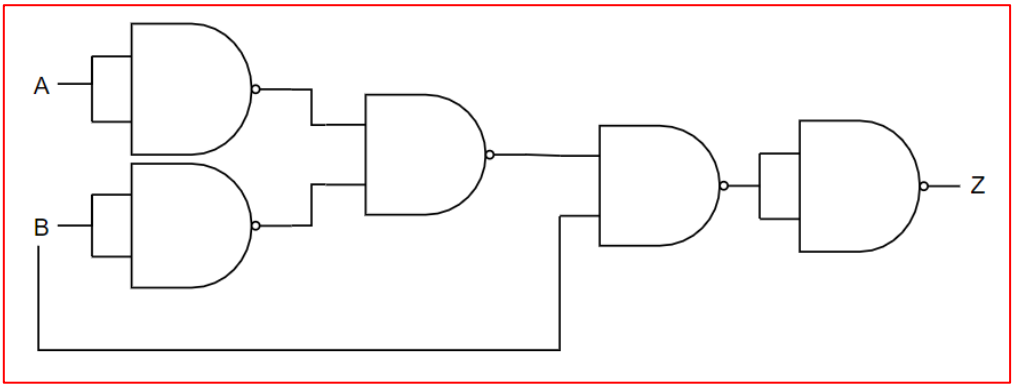
$D = \overline{C \cdot B}$

$Z = \overline{D}$

$Z = \overline{\overline{(A + B) \cdot B}}$

$Z = (A + B) \cdot B$

A	B	C	D	Z
0	0	1	0	0
0	1	0	1	1
1	0	0	1	0
1	1	0	1	1



16

(i) $Input\ conditions = 2^3 = 8$

(ii) To invert the signal from A and C

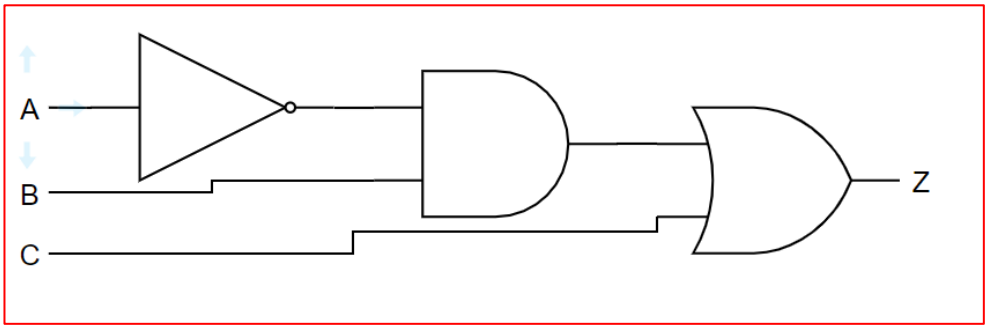
(iii)

A	B	C	D	E	F	G
0	0	0	1	1	1	0
0	1	1	1	0	0	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
1	0	0	0	1	1	0
1	1	1	0	0	1	1
1	0	1	0	0	1	1
1	1	0	0	1	1	0

(iv) Why is the design made up entirely of NAND gates?

a. Allow one IC chip to be used
b. Simplified design (advantages for electronic engineer)
c. Save money for manufacturer

(v)



17

a) $Z = (A \cdot \bar{B}) + (\bar{A} \cdot B)$

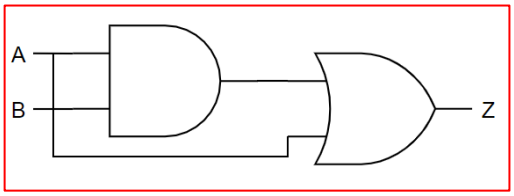
b) $Z = (\bar{A} \cdot B \cdot \bar{C}) + (\bar{A} \cdot B \cdot C) + (A \cdot \bar{B} \cdot C) + (A \cdot B \cdot C)$

c) $Z = (\bar{A} \cdot \bar{B}) + (A \cdot \bar{B}) = (\bar{A} + A) \cdot \bar{B}$

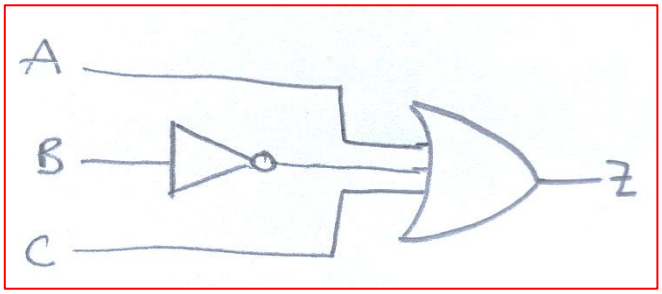
d) $Z = (\bar{A} \cdot \bar{B} \cdot \bar{C}) + (\bar{A} \cdot B \cdot \bar{C}) + (A \cdot B \cdot \bar{C}) = \bar{C}(\bar{A} \cdot \bar{B}) + (\bar{A} \cdot B) + (A \cdot B)$

18

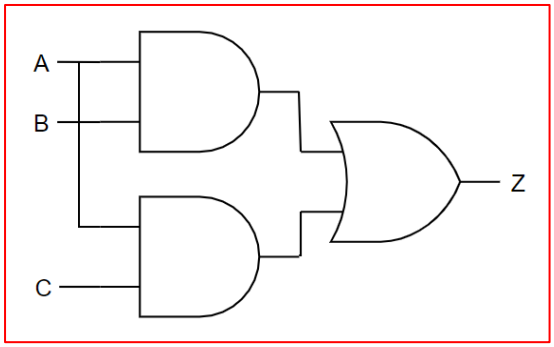
a)



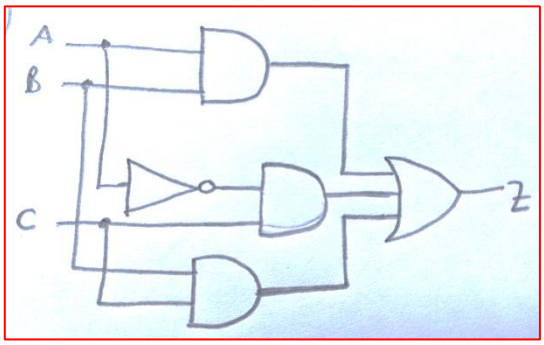
b)



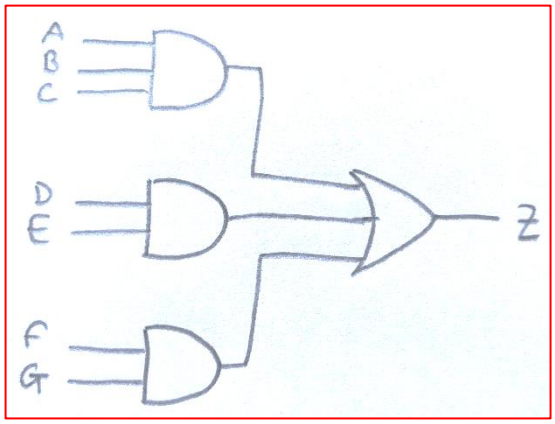
c)



d)

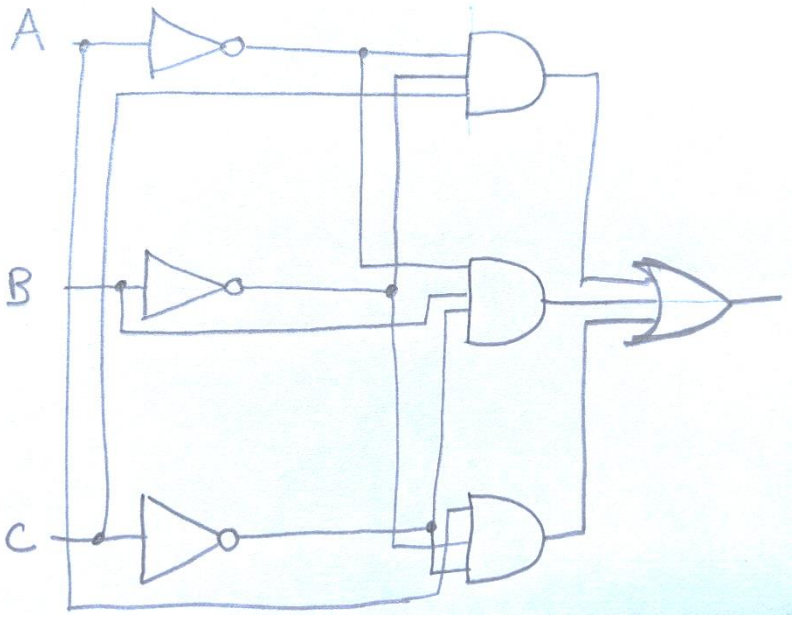


e)



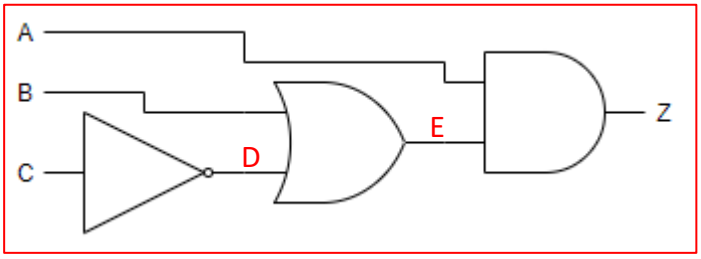
19

$$Z = (\bar{A} \cdot \bar{B} \cdot C) + (\bar{A} \cdot B \cdot \bar{C}) + (A \cdot \bar{B} \cdot \bar{C})$$



20

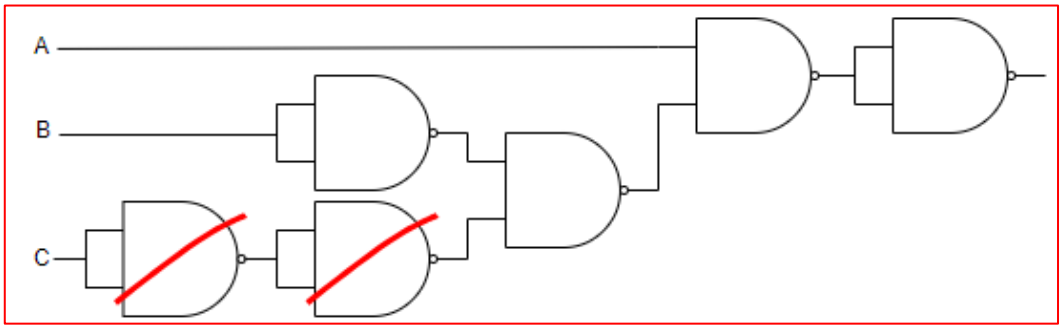
(i) Power Switch A, 1 = closed, 0 = open
Pressure Switch B, 1 = closed, 0 = open
Window switch C, 1 = closed, 0 = open
Output to bell Z



(ii)

A	B	C	D	E	Z
0	0	0	1	1	0
0	0	1	0	0	0
0	1	0	1	1	0
0	1	1	0	1	0
1	0	0	1	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	1	1

(iii)



21 Students must design the circuit using every step described in the method, to show they have understood every stage

Switch A – A
Switch B – B
Green light – G
Red light – R

Step 1:
Green light on if: *A is 1 and B is 1*
Red light on if: *A is 1 and B is 0* OR *A is 0 and B is 1*

Step 2:

Inputs		Green Light	Red Light
A	B	G	R
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Step 3:

Inputs		Green Light	Red Light
A	B	G	R
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$\bar{A} \cdot B$

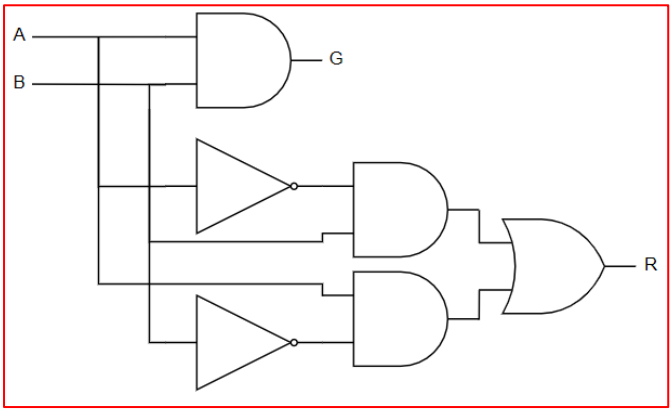
$A \cdot \bar{B}$

$A \cdot B$

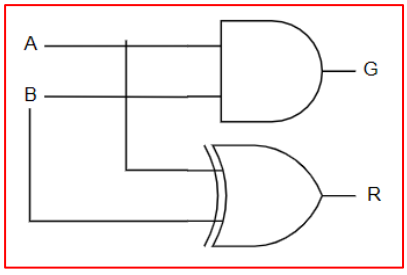
Step 4: $G = A \cdot B$ AND $R = \overline{(\bar{A} \cdot B)} + (A \cdot \bar{B})$

$R = A \oplus B$

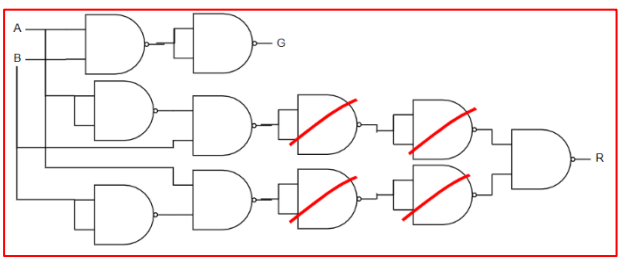
Step 5:



OR



Step 6:



Students are expected to redraw simplified answer. Students could also use NAND XOR gate config for a more efficient answer, as it uses 1 less gate.

22

Step 1:

X is on if: *A is 1 and B is 0 OR A is 0 and B is 1*

Y is on if: *A is 1 and B is 1*

Step 2:

A	B	X	Y
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Step 3:

A	B	X	Y
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

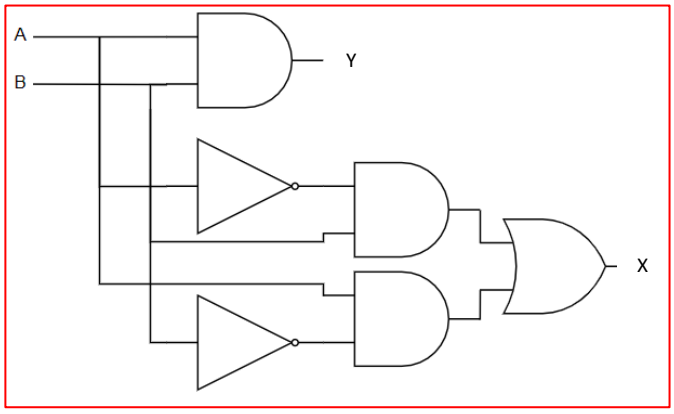
$\bar{A} \cdot B$

$A \cdot \bar{B}$

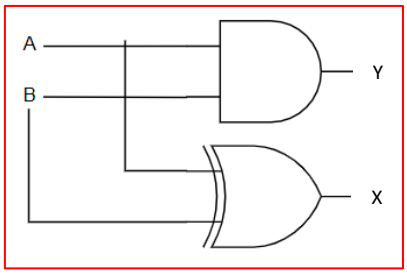
$A \cdot B$

Step 4: $X = (\overline{A} \cdot B) + (A \cdot \bar{B})$ OR $X = A \oplus B$ AND $Y = A \cdot B$

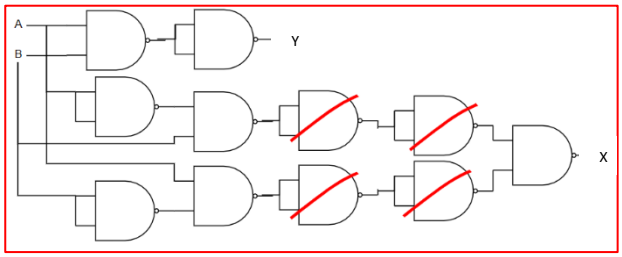
Step 5:



OR



Step 6:



Students are expected to redraw simplified answer. Students could also use NAND XOR gate config for a more efficient answer, as it uses 1 less gate.

23

Pressure switch - A, 1 = closed, 0 = open

LDR - B, 1 = active, 0 = inactive

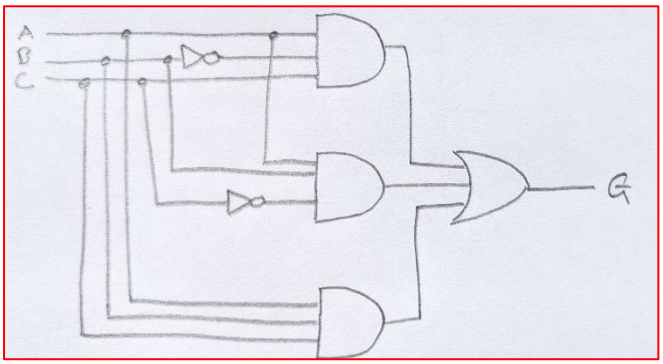
Key switch - A, 1 = operated, 0 = not operated

Garage door – G

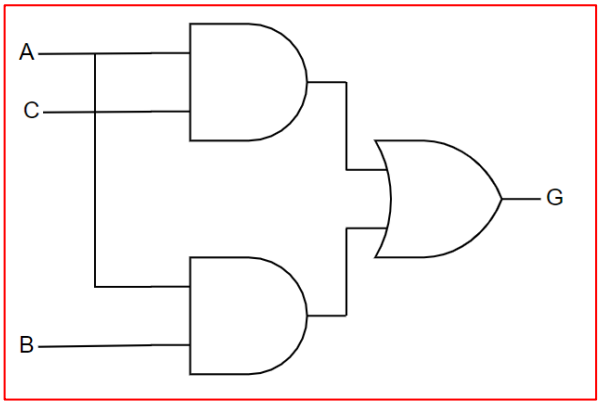
A	B	C	G
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

a)

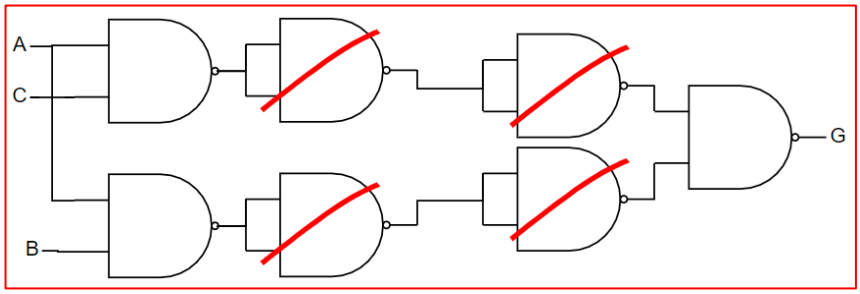
b) $G = (A \cdot \bar{B} \cdot C) + (A \cdot B \cdot \bar{C}) + (A \cdot B \cdot C)$



This circuit is correct, but will lead to a huge NAND equivalent, and doesn't answer the question as it does not use the least number of logic gates. Students should spot the following simplification:



c)



24

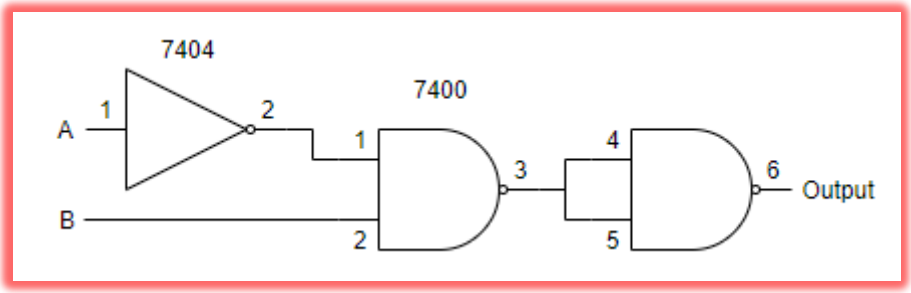
a) 7432 – OR gate
7404 – NOT gate
7400 – NAND gate

a) 7404 – NOT gate
7427 0 NOR gate

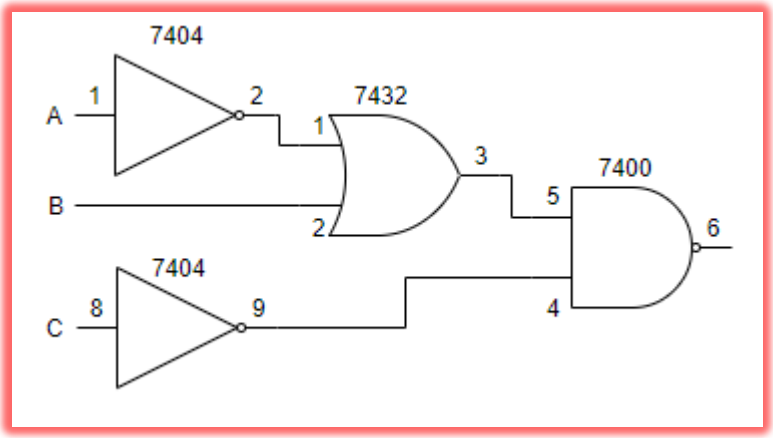
b) 7404 – NOT gate
7432 – OR gate
7400 – NAND gate

25

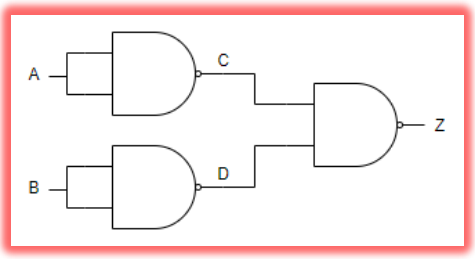
a)



b)



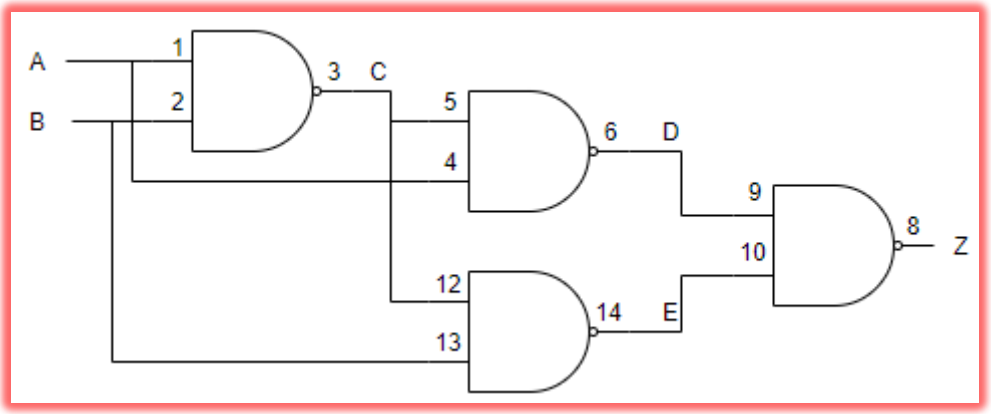
26



A	B	Z
0	0	0
0	1	1
1	0	1
1	1	1

This could be replaced by an OR gate

27



A	B	C	D	E	Z
0	0	1	1	1	0
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	1	1	0

This could be replaced by a XOR gate

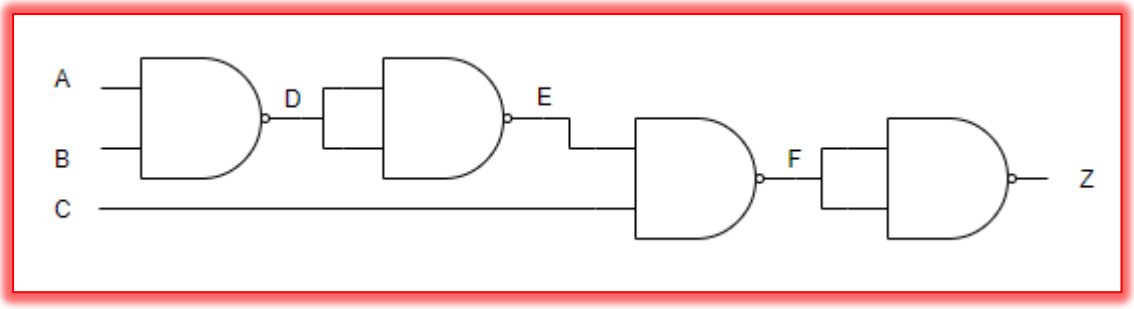
28

A	B	C	D	Z
0	0	1	1	0
0	1	1	0	1
1	0	0	1	1
1	1	0	0	1

This could be replaced by an OR gate

29

a)



i.

A	B	C	D	E	F	Z
0	0	0	1	0	1	0
0	0	1	1	0	1	0
0	1	0	1	0	1	0
0	1	1	1	0	1	0
1	0	0	1	0	1	0
1	0	1	1	0	1	0
1	1	0	0	1	1	0
1	1	1	0	1	0	1

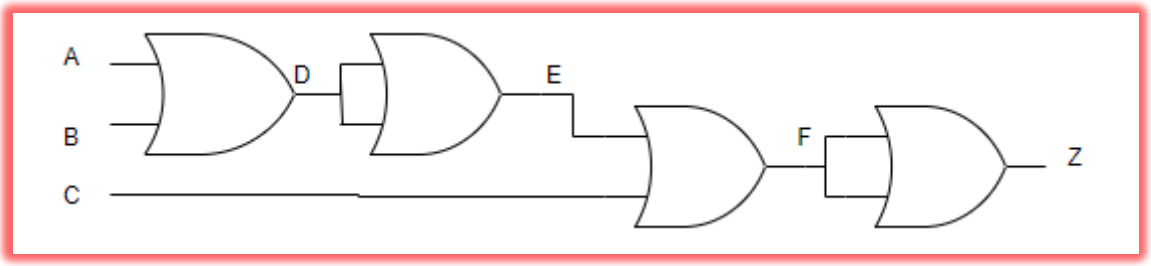
ii. Student should use Yenka to complete the table as above

iii. $Z = A \cdot B \cdot C$

This circuit could be replaced by a 3-input AND gate

b)

i.



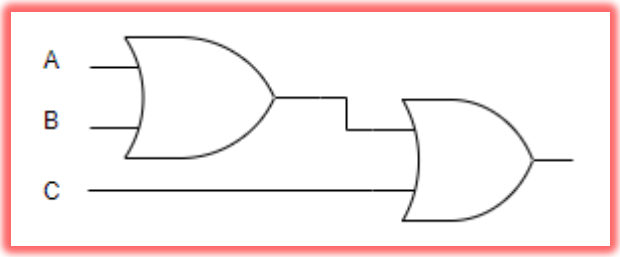
ii.

A	B	C	D	E	F	Z
0	0	0	0	0	0	0
0	0	1	0	0	1	1
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	1	1	1	1
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

Boolean expression:

$Z = A + B + C$

iii. This circuit could be replaced by a three input OR gate. Simplified logic diagram:



Exam style practice questions

1. The operation of an industrial stamping machine is controlled by a logic system with inputs and output as shown in the following table.

INPUTS	OUTPUT
L (= 1 when left-hand button is pressed)	S (= 1 to operate stamping machine)
R (= 1 when right-hand button is pressed)	
G (= 1 when guard is in position)	

The logic system shown in Figure Q1(*a*) ensures that the machine only operates when the guard is in place and either button is pressed.

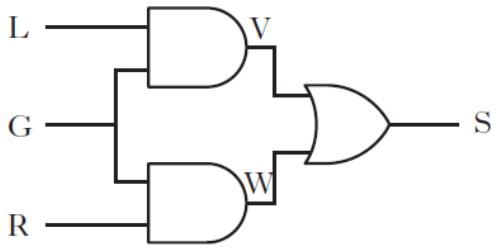


Figure Q1(*a*)

(*a*) Write a Boolean expression for the output S in terms of the inputs L, G and R. **3**

(*b*) Using only two-input NAND gates, draw a logic system equivalent to the logic system shown in Figure Q1(*a*). Delete any redundant gates. **4**

The logic system shown in Figure Q1(*b*) was designed to provide the same logic function as the logic system shown in Figure Q1(*a*).

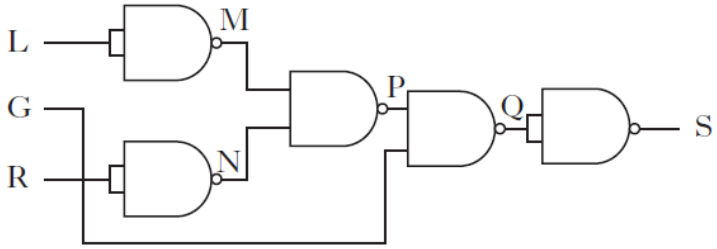
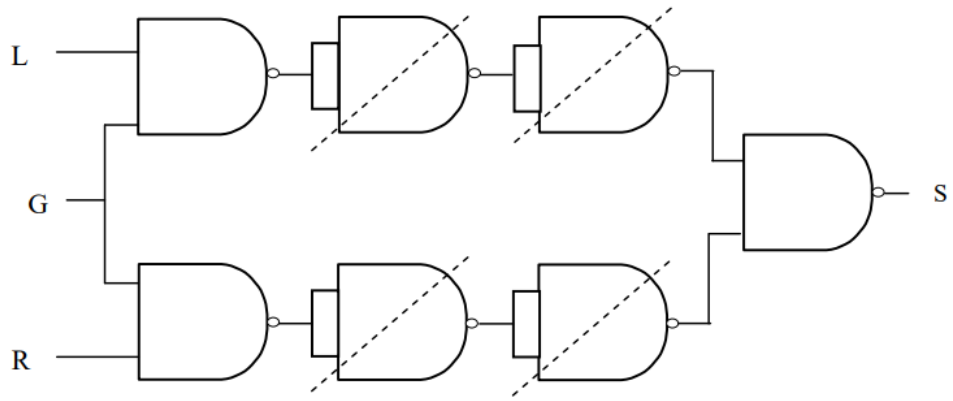
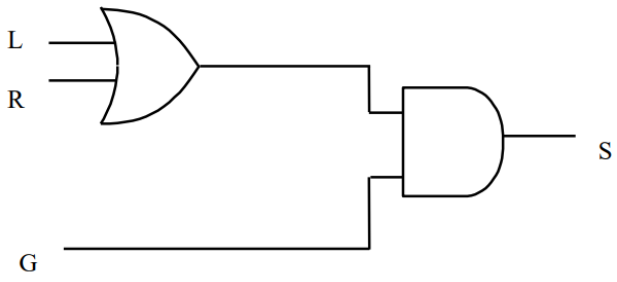


Figure Q1(*b*)

(*c*) Draw and complete a truth table for the logic system shown in Figure Q1(*b*). Include the intermediate logic values M, N, P & Q. **5**

(*d*) Draw a logic system equivalent to that shown in Figure Q1(*b*) using only two-input AND and OR gates. **3**

Answer:

Question		Mark Allocation		Marks	
1.	a	S = (L.G) + (R.G) (brackets not necessary) $S = (\bar{L}.G.R) + (L.G.\bar{R}) + (L.G.R)$ S = (L + R) . G	AND functions 1 each OR function 1 Each statement 1 L+R 1 .G 1 ( ) 1	3	**3**
	b		Two AND equivalents @ 1 each OR equivalent deleting redundant gates	2 1 1	**4**
	c				**5**
	d		correct OR correct AND correct interconnections	1 1 1	**3** **(15)**

L	G	R	M	N	P	Q	S
0	0	0	1	1	0	1	0
0	0	1	1	0	1	1	0
0	1	0	1	1	0	1	0
0	1	1	1	0	1	0	1
1	0	0	0	1	1	1	0
1	0	1	0	0	1	1	0
1	1	0	0	1	1	0	1
1	1	1	0	0	1	0	1

1 each fully correct column for M,N,P,Q and S

2. The tumble dryer shown in Figure Q1 vents warm air back into the house through a filter, and collects moisture from the clothes in a water tank. An air flow sensor detects when the filter needs to be cleaned.



Figure Q1

A logic system controls a warning buzzer. If the water tank is full (W=1) or the air flow is too low (A=0) when the start button is pressed (S=1), then the buzzer sounds (B=1).

(*a*) Draw a truth table for the output, B, in terms of the inputs S, W and A. **3**

(*b*) Write a Boolean expression for B in terms of S, W and A. **4**

(*c*) Draw a logic diagram for the control system, constructed from AND, OR and NOT gates. **3**

(*d*) Draw a logic diagram for the control system, using only NAND gates.
Simplify where appropriate. **4**

(14)

Answer:

Question	Mark Allocation	Marks

(a)

S	W	A	B
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

3 marks for all B output
2 marks 7 or 6 B correct
1 mark 5 or 4 B correct
0 marks 3 or less correct

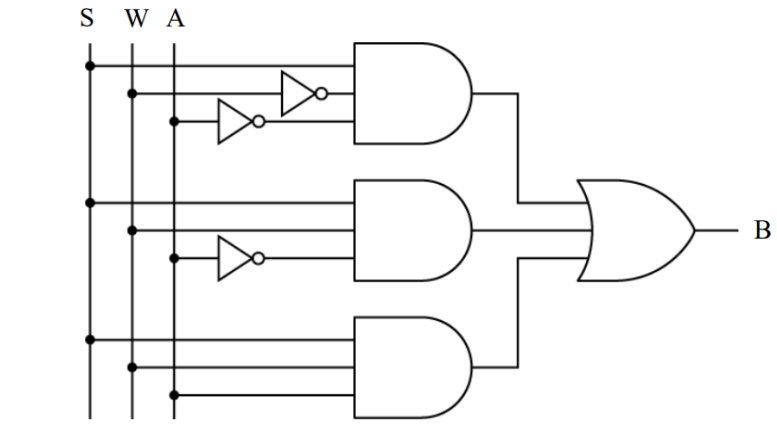
3

(b) $B = S.\overline{W}.\overline{A} + S.W.\overline{A} + S.W.A$ (1 mark) for each correct combination

OR

$B = S(W+A)$

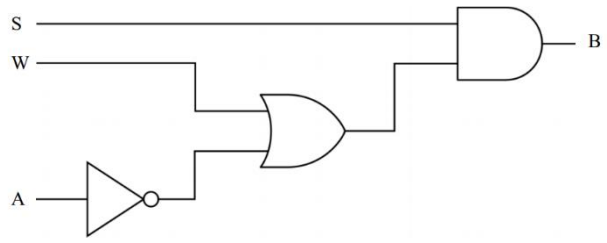
(c)



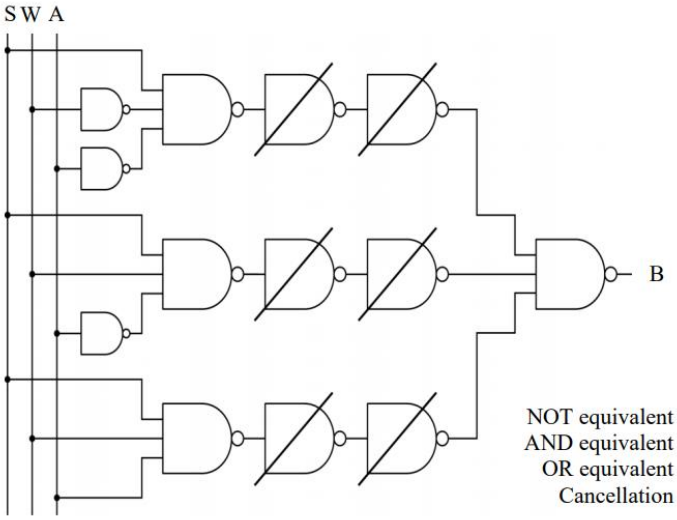
NOT 1
AND 1
OR 1

3

OR



(d)



NOT equivalent 1
AND equivalent 1
OR equivalent 1
Cancellation 1

4

(14)